

Section 1: Placement Profile and Learning Objectives

Placement Details

- **Company Name:** Lassonde School of Engineering, York University
- **Job Title:** Undergraduate Research Assistant
- **Supervisor Name:** Dr. Sunila Akbar
- **Supervisor Job Title:** Research Supervisor

Learning Objectives

This full-time placement focuses on developing advanced machine learning models under the project title "Adaptive AI for Recognizing Unknown Wireless Signals". The quantifiable objectives established for this term include:

1. **Critical Thinking and Problem Solving:** Compile evaluation metrics for newly implemented discrimination and triplet loss algorithms into a comprehensive results table, comparing signal to noise ratios (SNRs) and known/unknown data splits to integrate directly into an ongoing paper draft.
2. **Workplace Optimization:** Optimize the anomaly detection threshold for the Unknown Signal Bank to decrease the false positive rate by 10%, utilizing confusion matrices generated during incremental training phases to ensure high candidate selection purity.
3. **Academic Communication:** Write the "Numerical Results and Discussions" section of the manuscript, articulating the stability metrics of the dynamic adaptive threshold by the end of the term.
4. **Technical Collaboration:** Coordinate a comprehensive technical handover session in August, documenting the clustering implementation and codebase architecture to allow seamless subsequent modifications of the framework after my departure.

Section 2: Weekly Journal Entries and Narrative Log

Weeks 1 to 2: Literature Review, OpenICL Replication, and the Accidental Clustering Discovery

The first two weeks demanded an intensive literature review on open-set modulation classification, focusing on the foundational OpenICL research paper. The objective was to understand and replicate the core methodology. I successfully implemented the framework, but this included a significant caveat. During coding, I mistakenly implemented a localized clustering mechanism under the incorrect assumption that the original paper utilized that exact technique.

Diagnosing this discrepancy led to a major conceptual breakthrough. Prior open-set architectures frequently overfit when mapping unknown signals, attempting to establish an arbitrary Feature Layer boundary that forced unrelated, distinct unknown signals to crowd together in the latent space. Clustering these unknown signals instead preserves their unique spatial distributions. Each

unknown modulation type naturally forms its own separate cluster in the feature layer rather than being squeezed into a single artificial boundary. This prevents latent space distortion, representing a novel engineering contribution not implemented by previous papers.

Weeks 3 to 4: Thinking of Improvements, Framework Setup, and EVT Integration

Following the initial replication, the next two weeks shifted toward brainstorming improvements and implementing Extreme Value Theory (EVT) alongside a batch all triplet loss function. The project relied on the hybrid DONet architecture, which splits feature extraction into a Classification Path and a Contrastive Path.

The goal was to utilize the dual loss function to simultaneously optimize both paths, forcing the network to train a singular, highly robust feature layer. The Classification Path enforces discrete boundaries for known signals, while the Contrastive Path forces sample embeddings to map logically based on structural similarities. During this period, I established a weekly routine of coding, running Compute Canada high-performance simulations, and meeting Dr. Sunila Akbar twice weekly to discuss progress.

Week 5: Algorithmic Refinement and the Removal of EVT

During the fifth week, the focus turned toward iterative optimization based on the benchmark data. While Extreme Value Theory provided a mathematical baseline for statistical boundary estimation, empirical results indicated it introduced unnecessary complexity without yielding precision in dynamic environments. Consequently, I strategically removed EVT from the architecture altogether. Instead, I focused on directly improving the batch all triplet loss function. By tuning margin hyperparameters and altering how the network processes positive and negative signal pairs, I streamlined the architecture for superior classification accuracy.

Weeks 6 to 7: Benchmarking and Manuscript Integration

The subsequent two weeks were dedicated to rigorous benchmarking. I ran extensive simulations across various signal to noise ratios (SNRs) and data splits on Compute Canada. Throughout these weeks, as benchmark results finished computing, I systematically added the finalized data comparison tables and figures directly into our active research paper draft to keep the manuscript updated.

Week 8: Bug Fixing and the 10% F1-Score Breakthrough

The eighth week brought a critical technical breakthrough. While reviewing the codebase and analyzing performance anomalies across specific sub-datasets, I discovered a hidden bug in the batch all triplet loss implementation. Fixing this bug completely transformed the model's behavior. By correcting the loss calculation, the framework achieved a definitive 10% improvement in its F1-score. This explicit quantification of an open-set accuracy rise, achieved without losing classification accuracy in closed-set known signal distributions, represented a major project milestone.

Week 9 (Current Phase): Testing Advanced Clustering Algorithms

Currently, I have shifted testing to different clustering algorithms to improve accuracy. The model previously relied on a k -means baseline with silhouette scoring, which struggles with irregular boundaries. I am transitioning to Density-Based Spatial Clustering of Applications with Noise (DBSCAN). Because dynamic wireless signals are inherently noisy, DBSCAN evaluates localized density pockets, effectively isolating true clusters from outlier noise. This phase involves heavy literature review to find supporting density-based techniques.

Section 3: Met Objectives, Retrospective, and Growth Analysis

Synthesis of LE/COOP 2100 Course Concepts (Retell, Relate, Reflect)

This placement required an immediate, practical synthesis of the professional development frameworks taught in LE/COOP 2100. Rather than simply logging my daily tasks, the course required me to apply the Retell, Relate, Reflect model to evaluate my professional evolution.

First, my experience resolving the hidden bug in the batch all triplet loss directly connected to the Growth Mindset explored in Module 3. In a fixed mindset, encountering a significant technical roadblock could be interpreted as a failure. However, by actively applying a growth mindset, I viewed this challenge as a springboard for development. Diagnosing the bug required extensive effort and learning from failures, ultimately leading to the 10% F1-score breakthrough.

Second, coordinating with Dr. Akbar required the application of Emotional Intelligence (EQ) from Module 2. A core domain of EQ is Self Management, involving the ability to control disruptive impulses and suspend judgment. When Dr. Akbar noted that I needed to explain my technical methods more clearly, my initial internal reaction could have been defensive. Instead, applying Self Management allowed me to process this critique objectively. I actively adapted my communication style by creating dedicated presentations and drawing simplified visuals. Using basic tools like MS Paint, I drafted visual representations to map out my abstract latent space operations during our meetings. This initiative drastically improved our communication and demonstrated adaptability.

Finally, managing my professional trajectory forced me to synthesize the Personal SWOT Analysis from Module 5 with the SMART Goal framework. Through the SWOT Analysis, I identified a specific weakness regarding my ability to translate abstract mathematical operations into accessible verbal explanations. I mapped this weakness directly to the Undergraduate Competency of Effective Oral and Written Communication. By focusing heavily on visual aids and structured presentations, I am actively transforming this weakness into an opportunity for career development.

Midterm Evaluation of Learning Objectives: Meeting and Exceeding Expectations

At this halfway point of the 16 week placement, an honest assessment reveals that I am consistently meeting, and in several technical areas exceeding, my initial learning objectives.

Regarding my technical goals, I have significantly exceeded expectations. The objective to optimize the anomaly detection threshold and decrease the false positive rate by 10% was successfully

realized much earlier than anticipated through the batch all triplet loss bug fix. I successfully compiled the evaluation metrics and advanced the codebase further than expected by pioneering the novel unknown signal clustering approach.

Regarding my communication and collaboration goals, these remain actively in progress. I continuously draft the "Numerical Results and Discussions" section as new DBSCAN benchmark results are generated. Although final technical handovers and manuscript revision cycles will occur in August, I proactively comment code and prepare documentation to ensure I meet these collaboration targets.

Personal and Professional Development

Professionally, this research term has provided deep insight into the collaborative nature of modern science. The entire philosophy of academic research relies on standing upon the structural foundations built by previous engineers. This placement has required seamless collaboration at multiple levels: working directly with Dr. Akbar to co-develop algorithmic ideas, and relying on external researchers to obtain authentic datasets.

Over the past two months, my technical writing and verbal communication have improved significantly. Learning to articulate exactly how the Classification and Contrastive paths jointly optimize the feature layer, and supporting those explanations with custom visuals, has built immense confidence. Ultimately, the reflective frameworks established in LE/COOP 2100 provided the exact tools necessary to navigate workplace challenges, accept constructive feedback, and systematically exceed my technical benchmarks. This newfound ability to break down complex problems will directly translate to my upcoming academic responsibilities, where I will serve as a supportive Peer Mentor delivering clear mathematical explanations to junior engineering students.